

Chapter-01 exercise solution

1. Which of the following correctly explains a network characteristic?
 - ✓ Reliability is often measured as a probability of failure or as the mean time between failures (MTBF).
2. What are two functions of a router?
 - ✓ It connects multiple IP networks
 - ✓ It determines the best path to send packets.
3. Which two statements correctly describe the concepts of administrative distance and metric?
 - ✓ Administrative distance refers to the trustworthiness of a particular route.
 - ✓ Routes with the smallest metric to a destination indicate the best path.
4. For packets to be sent to a remote destination, what three pieces of information must be configured on a host?
 - ✓ Default gateway
 - ✓ IP address
 - ✓ Subnet mask
5. What is a characteristic of an IPv4 loopback interface on a Cisco IOS router?

It is a logical interface internal to the router.
6. What two pieces of information are displayed in the output of the show ip interface brief command?
 - ✓ Interface descriptions
 - ✓ IP addresses
7. A packet moves from a host on one network to a device on a remote network within the same company. In most cases, which two items remain unchanged during the transfer of the packet from source to destination?
 - ✓ Destination IP address.
 - ✓ Source IP address
8. Which two items are used by a host device when performing an ANDing operation to determine whether a destination address is on the same local network? (Choose two.)
 - ✓ Destination IP address
 - ✓ Subnet mask
9. What will the router do with a packet that has a destination IP address of 192.168.12.227?
 - ✓ Send the packet out the Serial0/0/0 interface.
10. Which two parameters does EIGRP use as metrics to select the best path to reach a network?
 - ✓ Bandwidth
 - ✓ Delay
11. What route would have the lowest administrative distance?
 - ✓ A directly connected network.

12. Consider the following routing table entry for R1: D 10.1.1.0/24 [90/2170112] via 10.2.1.1, 00:00:05, Serial0/0/0 What is the significance of the Serial0/0/0?
- ✓ It is the interface on R1 used to send data that is destined for 10.1.1.0/24.
13. A network administrator issues the show ipv6 route command on R1. What two conclusions can be drawn from the routing table?
- ✓ Packets destined for the network 2001:DB8:ACAD:2::/64 will be forwarded through G0/1.
 - ✓ R1 does not have any remote network routes.
14. A network administrator configures interface G0/0 on R1 with the ip address 172.16.1.254 255.255.255.0 command. However, when the administrator issues the show ip route command, the routing table does not show the directly connected network. What is the possible cause of the problem?
- ✓ Interface G0/0 has not been activated.
15. A network administrator configures a router using the command ip route 0.0.0.0 0.0.0.0 209.165.200.226. What is the purpose of this command?
- ✓ To provide a route to forward packets for which there is no route in the routing table.
16. What are two common types of static routes in routing tables?
- ✓ A default static route and
 - ✓ A static route to a specific network
17. What command will enable a router to begin sending messages that allow it to configure a link-local address without using an IPv6 DHCP server?
- ✓ The ipv6 unicast-routing command.